

development platform, it gives you the capability to simulate your target directly in a PC.

The good and the bad

The x86 platform was widely adopted for hybrid PC technology-based applications rather than traditional embedded applications. Listing certain merits of working with x86 platform, Trivedi says, "The initial set-up time for software development is quick. Also, the software developed on x86 are scalable, compatible and easy to interface/port to legacy applications. It also significantly reduces the dependency on dedicated hardware." However, PCB footprint and power requirement need to be evaluated well before choosing x86 for any application.

Performance per watt is the consideration. The demand for more features and functions for less power is increasing with every innovation in the embedded space. Although x86 delivers the greatest performance, when it comes to power efficiency it becomes the least preferred architecture. In the desktop world the push is to achieve more performance, while in the embedded space performance per watt counts.

According to some hardware engineers, ARM's dominance in small electronic products is simply because of its high power efficiency. Typically, an ARM-based system uses as little as 2 watts, whereas a fully optimised x86 solution such as Atom uses 5-6 watts.

Hongyu You, senior software developer, Infinera, says, "A battery-powered device with an 8051 is different from a set-top box, network switch or storage array which may be plugged into a power outlet in the wall and may require much more processing power."

Although there are ways and means to manage power, "The power consumption and thermal efficiency of x86 make it better suited for continuous AC-powered applications with forced-air or liquid cooling rather than mobile applications," says Mike

Anderson, chief scientist at The PTR Group.

According to Anderson, Intel's Atom seems to be moving in the right direction, but it's a long way before it can come close to touching the power efficiency or MIPs/watt ratio found in many ARM variants. There are also cooling issues associated with x86, as for embedded systems developers prefer the processor which does not require external cooling.

Standardisation and integration. Another advantage of x86 architecture is that the standard commercial software is often designed to run on x86-based CPUs. This is also true for many plug-and-play gadgets like printers, keyboards and monitors.

Trivedi shares, "x86 scores over other platforms when the application is very specific and tightly integrated with legacy counterpart." If you are looking for easy integration and a wide range of peripherals like lots of GPIO, PCI, UART, analogue-to-digital and digital-to-analogue converters, and storage, the x86 solution is a good choice.

Also, x86 architecture is highly scalable. That means if you have designed an embedded system based on a small x86 processor like an Atom,

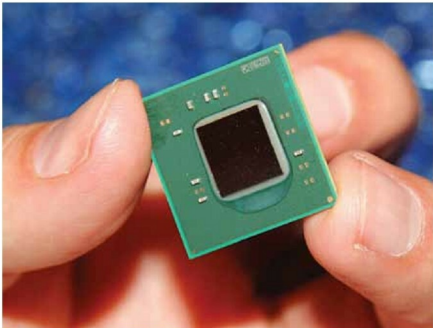
you can easily replace it with a more powerful processor like Core Duo without much of redevelopment.

Since every processor in the ARM family is individually developed with specific features, it cannot be scaled down or up. For an ARM-based solution, you need to know the exact need of the embedded solution.

Coming to a real-time embedded application, there is hardly any instance when you have to upgrade just the processor of the application. By its very definition, embedded applications are developed to perform dedicated tasks.

Booting x86-based system. x86 usually comes with basic input/output system (BIOS). Leonid Rosenboim, an embedded systems entrepreneur, says, "BIOS is very useful for booting, but it is the software that you do not control, and may need to change when it is undesirable. This could prevent you from having full control over some resources. Some types of BIOS could actually continue to run in the background along with your RTOS and application, causing occasional mysterious deadline slips."

In x86, the BIOS can be replaced with a less expensive, lighter-weight



Intel Atom processor